

Question 1

[10]

Design of 4:16 decoder: The decoder has four inputs A_3, A_2, A_1 and A_0 and its complements $\bar{A}_3, \bar{A}_2, \bar{A}_1$ and $\bar{A}_0$. These 8 inputs are passed through 8 inverters having 10 units of input capacitance each, thus forming 1st stage of decoder. The remaining stages of the decoder has following different combinations to produce final decoder outputs. Assume that each decoder is driving 100 units of output capacitance. For each combination, calculate minimum possible normalized delay. Assume that drive strength of PMOS is half of NMOS for the same size. (Hint: Be careful about branching effort)

Combination	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7	Minimum total normalized delay
1	NAND4	INV	-	-	-	-	26.52τ
2	NAND2	NOR2	-	-	-	-	47.51τ
3	INV	NAND4	INV	-	-	-	23.92τ
4	NAND4	INV	INV	INV	-	-	23.85τ
5	NAND2	INV	NAND2	INV	INV	INV	33.71τ

Important Steps:

$$\textcircled{1} : G = 1 \times 2 \times 1 = 2$$

$$B = 1 \times 16 \times 1 = 16$$

$$H = \frac{100}{10} = 10$$

$$GBH = 320$$

$$B = 6.83$$

$$P = 1 + 4 + 1 = 6$$

$$D = NB + P$$

$$= 26.519$$

$$\textcircled{2} : G = 1 \times \frac{4}{3} \times \frac{5}{3} = \frac{20}{9}$$

$$B = 1 \times 8 \times 16 = 128$$

$$H = \frac{100}{10} = 10$$

$$GBH = 2844.44$$

$$B = 14.168$$

$$P = 1 + 2 + 2 = 5$$

$$D = NB + P$$

$$= 47.51$$

$$\textcircled{3} : G = 1 \times 1 \times 2 \times 1 = 2$$

$$B = 1 \times 1 \times 16 \times 1 = 16$$

$$H = \frac{100}{10} = 10$$

$$GBH = 2 \times 16 \times 10 = 320$$

$$B = 4.229$$

$$P = 1 + 1 + 4 + 1 = 7$$

$$D = NB + P$$

$$= 23.92$$

$$\textcircled{4} : G = 1 \times 2 \times 1 \times 1 \times 1$$

$$B = 1 \times 16 \times 1 \times 1 \times 1$$

$$H = \frac{100}{10}$$

$$GBH = 320$$

$$B = 3.17$$

$$P = 1 + 4 + 1 + 1 + 1 = 8$$

$$D = NB + P$$

$$= 23.85$$

$$\textcircled{5} : G = 1 \times \frac{4}{3} \times 1 \times \frac{4}{3} \times 1 \times 1 \times 1$$

$$B = 1 \times 8 \times 1 \times 16 \times 1 \times 1 \times 1$$

$$H = \frac{100}{10} = 10$$

$$GBH = 6826.67$$

$$B = 3.53$$

$$P = NB + P$$

$$= 33.71$$

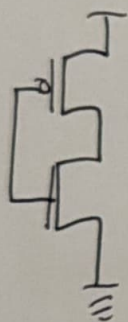
$$P = 1 + 2 + 1 + 2 + 1 + 1 + 1$$

$$= 9$$

Question 2[18]

Assuming the drive strength of PMOS is $1/3$ times of NMOS for the same size, find the size of NMOS and PMOS in pseudo NMOS inverter where pull up strength is $1/6$ times that of NMOS pull-down network. Also find the parameters specified in the table below for gates designed using same pseudo NMOS logic. Write values as ratio of 2 integers.

Gate	Size of NMOS	Size of PMOS	g_u	g_d	P_u	P_d
pseudo NMOS inverter	1	$1/2$	$9/4$	$3/8$	$9/4$	$3/8$
pseudo NMOS NOR2	1	1	$12/7$	$2/7$	$9/4$	$3/8$
pseudo NMOS NAND2	2	$1/2$	$3/1$	$1/2$	$9/4$	$3/8$

Important Steps:

$$\frac{3R}{K} = GR$$

$$K = 1/2$$

$$\tau = \frac{3}{2} C \times 6R$$

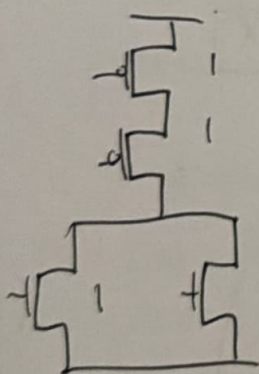
$$= 9RC$$

$$\frac{3}{2} C \times R$$

$$= \frac{3}{2} RC$$

$$g_u = \frac{1+1/2}{1/2+1/6} = 2.25$$

$$g_d = \frac{1+1/2}{1+3} = \frac{3}{8}$$



$$\frac{3R}{K} + \frac{3R}{K} = GR$$

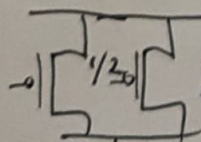
$$K = 1$$

$$g_u = \frac{1+1}{1+1/6} = \frac{12}{7}$$

$$P_u = \frac{1+H1}{1+\frac{1}{6}+\frac{1}{6}} = \frac{9}{4}$$

$$g_d = \frac{1+1}{1+6} = \frac{2}{7}$$

$$P_d = \frac{1+1+1}{1+1+6} = \frac{3}{8}$$



$$\frac{1}{2} \frac{3R}{K} = GR, K = 1/2$$

$$g_u = \frac{1/2+2}{1/2+1/3} = 3$$

$$P_u = \frac{1/2+1/2+2}{1/2+1/2+1/3} = \frac{9}{4}$$

$$g_d = \frac{1/2+2}{3+2} = \frac{1}{2}$$

$$P_d = \frac{1/2+1/2+2}{2+3+3} = \frac{3}{8}$$

$$R/K = R$$

$$K = 1$$

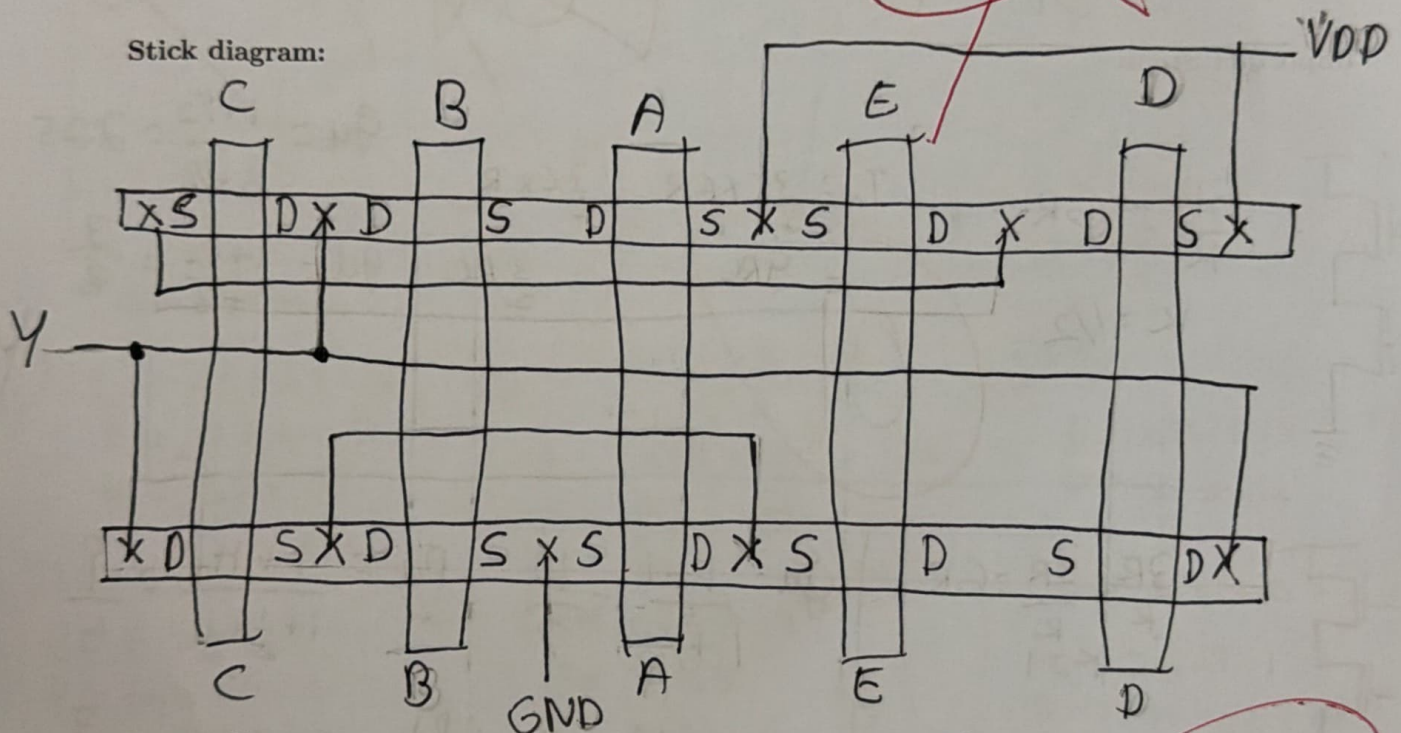
Question 3[10]

Implement logic function $\overline{(A+B)}(C+DE)$ using CMOS technology such that the total diffusion capacitance is minimized and size the PMOS and NMOS such that worst case drive for pull-up and pull-down is equal. Draw stick diagram using Euler's path common to PMOS and NMOS network. Draw circuit diagram clearly showing size of each transistor and total capacitance at each node as per stick diagram. Assume that in NMOS network D is close to output and in PMOS network, B is close to output. Also for same drive, PMOS to NMOS size is 2:1. Assume same size for shared contacted and shared un-contacted diffusion.

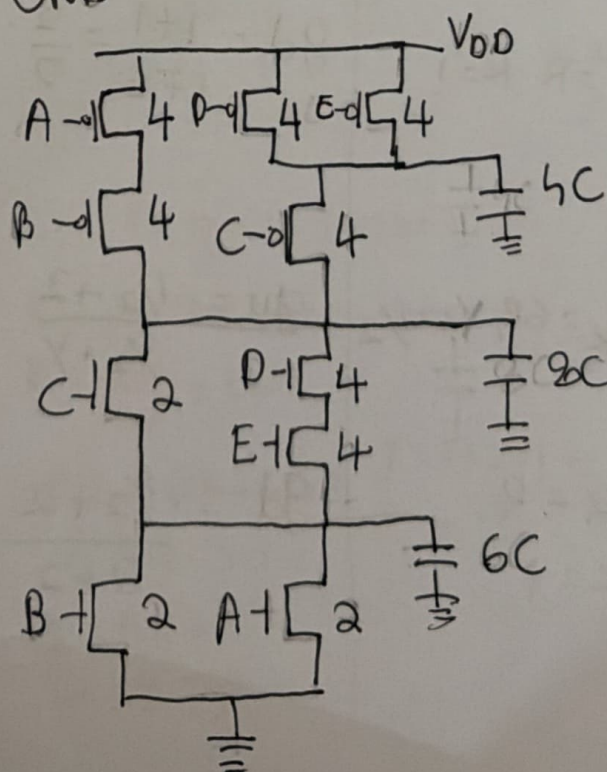
Euler's common path for NMOS and PMOS network:

~~CBAED~~

Stick diagram:



Circuit diagram:



01

Question 4

For the circuit shown below, find setup and hold time at inputs A, B, E with respect to clock CLK. [14]

Also find maximum operating frequency of the circuit.

Component	Flip-flop clk-to-Q	XOR	XNOR	NAND	OR	NOR	BUFFER
Minimum Delay	5ns	5ns	4ns	2ns	4ns	3ns	1ns
Maximum delay	8ns	8ns	7ns	4ns	6ns	5ns	2ns

Setup and hold time of Flip-flops is 4 ns and 3 ns respectively.

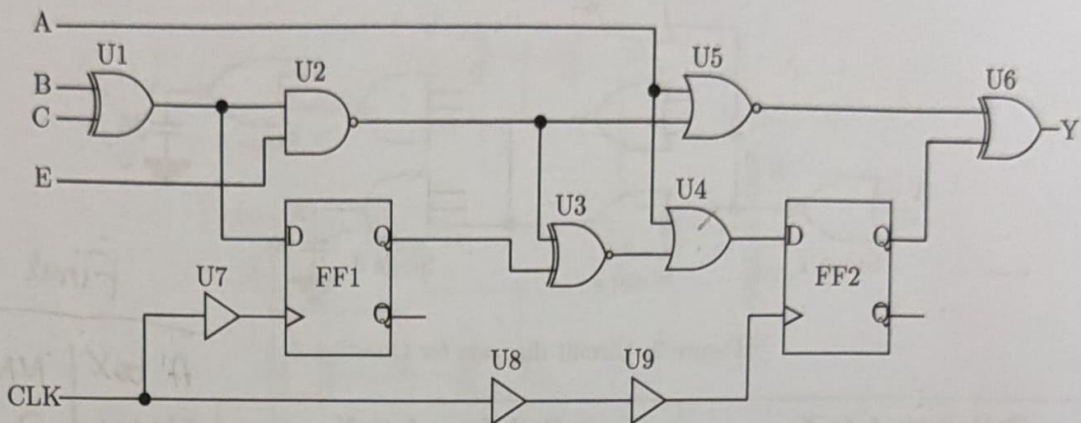


Figure 1: Circuit diagram for Question 4

Inputs	Setup time	Hold time
A	7ns	4ns
B	26ns	0ns
E	19ns	0ns
Maximum operating frequency = 40MHz 35.71MHz		

Max path:

$$B-U1-U2-U5-U6 = 25ns$$

$$\therefore Max f = 0.046Hz = 40MHz$$

$$B-U1-U2-U3-U4-FF2$$

$$= 24 + 4 = 28ns$$

$$f = \frac{1}{28} =$$

$$35.71MHz$$

Important Steps:

A) Data path: A-U4-FF2 CLK path: CLK-U8-U9-FF2
 max = 3ns min = 3ns max = 4ns min = 2ns
 Setup = 5 + 4 - 2 = 7ns hold = 4 + 3 - 3 = 4ns

B) Datapath: B-U1-FF1 - min = 5ns
 B-U1-U2-U3-U4-FF2 - max = 24ns
 Setup = 24 + 4 - 2 = 26ns hold = 2 + 3 - 5 = 0

E) Datapath: E-U2-U3-U4-FF2
 max = 17ns min = 9ns
 Setup = 17 + 4 - 2 = 19ns hold = 4 + 3 - 9 = -2 = 0

Question 5 [13]

In the circuit diagram shown below the input A has capacitance of 14λ of transistor width while output X and Y have capacitance of 60λ and 75λ of transistor widths respectively. If in a given technology, PMOS to NMOS ratio is 2:1 for same drive strength and FO4 inverter delay is 30 ps, find the minimum delay (not normalized) for path from A to X and for path from A to Y. Also find the size of NMOS and PMOS in each stage on both the paths in order to achieve this minimum delay. Each path has to be examined independently.

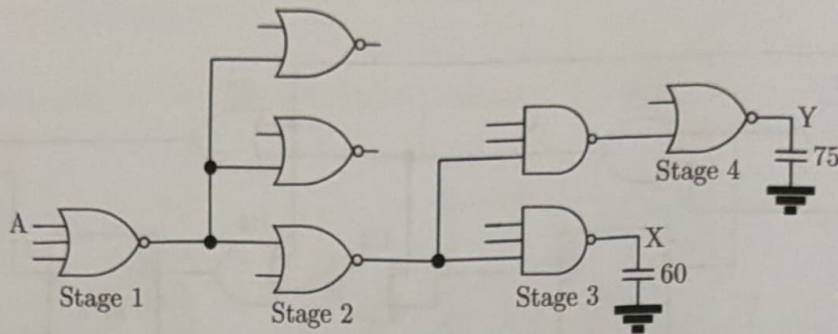


Figure 2: Circuit diagram for Question 5

Final Answer

Path from A to X			Path from A to Y		
Minimum Delay from input A to output X	147.06ps (3)		Minimum Delay from input A to output Y	163.6ps (3)	
	SIZE			SIZE	
	NMOS	PMOS		NMOS	PMOS
Stage 1 (1)	0.33	2	Stage 1 (1)	0.33	2
Stage 2 (1)	1.1	4.4	Stage 2 (1)	0.86	3.45
Stage 3 (1)	10.9	7.27	Stage 3 (1)	6.71	4.47
			Stage 4 (1)	5.79	23.17

A to X	NMOS	PMOS
Stage 1	2	12
Stage 2	2.2	9.8
Stage 3	10.9	7.27

A to Y	NMOS	PMOS
Stage 1	2	12
Stage 2	1.72	6.91
Stage 3	6.71	4.47
Stage 4	5.79	23.17

Important Steps:

A-X

$$G = g_1 \times g_2 \times g_3 = \frac{7}{3} \times \frac{5}{3} \times \frac{5}{3} = \frac{125}{27}$$

$$B = 3 \times 2 = 6$$

$$H = \frac{60}{14} = \frac{30}{7}$$

$$F = GBH = 166.67$$

$$B = \sqrt[3]{166.67}$$

$$= 5.5$$

$$D = NB + P$$

$$= 24.51$$

$$\text{delay} = 24.51 \times \tau =$$

$$G = g_1 \times g_2 \times g_3 \times g_4 = \frac{7 \times 5 \times 5 \times 5}{3 \times 3 \times 3 \times 3} = \frac{875}{81}$$

$$B = 2 \times 3 = 6$$

$$H = \frac{75}{14}$$

$$F = GBH = 347.22$$

$$B = \sqrt[4]{347.22} = 4.3167$$

$$D = NB + P$$

$$= 27.266$$

$$\text{delay} =$$

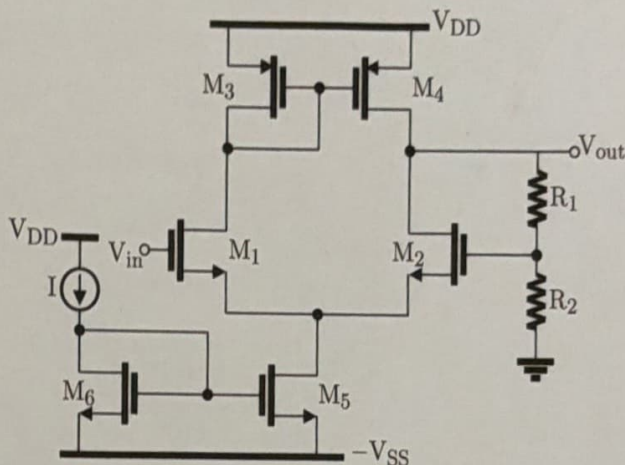
Question 6

[15]

Without feedback: Design an operational transconductance amplifier with differential gain of 80 V/V such that overdrive voltage of all transistors is same and $2(W/L)_6 = (W/L)_5$. $V_{DD} = V_{SS} = 0.9$ V, and power budget is 0.54 mW. What is the maximum output voltage swing? Calculate size of MOSFETs, maximum output voltage swing, $V_{in,CM,min}$ and $V_{in,CM,max}$.

Use $\mu_n C_{ox} = 300 \mu A/V^2$, $\lambda_n = 0.19 V^{-1}$ for $L = 1 \mu m$, $V_{th0,n} = 0.38$ V.

$\mu_p C_{ox} = 60 \mu A/V^2$, $\lambda_p = 0.057 V^{-1}$ for $L = 1 \mu m$, $|V_{th0,p}| = 0.5$ V.



With feedback: Design the feedback network such that $R_1 + R_2 = 20$ M Ω and ideal closed loop gain is 20 V/V. What is the closed loop gain with loading effect? What is output impedance without feedback with loading effect? What is output impedance with feedback including loading effect?

Size of M_1, M_2	Size of M_3, M_4	Size of M_5	Maximum output voltage swing	$V_{in,CM,min}$	$V_{in,CM,max}$
① 65.07	① 325.38	① 130.15	② 1.496V	0.48V	1.18V
Value of R_1	Value of R_2	Closed loop gain with loading effect	Output impedance without feedback with loading effect	Output impedance with feedback with loading effect	
19.25 MΩ	0.75 MΩ				

Important Steps:

$$I = \frac{0.54}{3 \times 1.8} = 100 \mu A$$

$$I = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_n V_{ov}^2$$

$$\frac{W}{L} = 65.07$$

$$A_v = g_m r_{o2} || r_{o4}$$

$$80 = \frac{2I}{V_{ov}} \times \frac{1}{\lambda (0.19 + 0.057)}$$

$$V_{ov} = 0.101 V$$

$$\text{Max swing} = V_{DD} - V_{ov,n} - (V_{ov,p} + V_{ov,2} - V_{SS})$$

$$= 1.496$$